

Project Title
Contract & Compliance Version-Intelligence Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Large organisations manage thousands of contracts that are revised repeatedly, making it difficult to track how clauses evolve and whether every regulatory obligation remains covered. Manual review at this scale is slow, inconsistent, and error-prone. This project presents a Contract & Compliance Version-Intelligence Platform that ingests a corpus of contracts and revisions and exposes a public API to reason about them programmatically. An inverted index over clause text supports fast clause search, while a sequence-alignment diff engine highlights changes between any two versions. Contract relatedness is found by computing pairwise clause-similarity scores and clustering contracts on the resulting graph. Obligation coverage is treated as a minimum set cover problem, using a greedy approximation to pick the smallest clause set satisfying a given obligation list. Reviewer assignment is modelled as bipartite matching, balancing workload against clause complexity. Together these components replace manual contract auditing with a fast, consistent, auditable process.

Introduction

As organisations grow, the number of contracts they hold and the number of times each one is revised grows far faster than the legal and compliance teams responsible for reviewing them. A single revision can add, remove, or reword clauses that affect regulatory obligations, and identifying exactly what changed between two versions of a lengthy contract is tedious to do by hand. Existing practice typically relies on reviewers reading contracts side by side, or on generic document-comparison tools that operate on raw text rather than on clauses as meaningful units, so a clause that has only been reworded is flagged as an unrelated change, and genuinely new obligations can be missed entirely.

These manual and text-level approaches also struggle to answer organisation-wide questions: which contracts are similar enough to be reviewed together, whether a given list of regulatory obligations is fully covered by the clauses already in force, and which reviewer should be assigned to which contract given their current workload. This project treats the problem as a set of well-defined algorithmic tasks, namely clause indexing and search, structured version diffing, similarity-based clustering, minimum set cover for obligation coverage, and assignment-based reviewer allocation, and exposes each of them through a single queryable platform.

Software and hardware requirements

Software:

- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python 3.8 or above
- **API Framework:** FastAPI or Flask, to expose the public REST API
- **Development Environment:** Google Colab
- **Core Libraries:** NetworkX (graph representation and graph algorithms), difflib (version diffing), scikit-learn (similarity and clustering utilities), SQLite3 (corpus storage)
- **Testing Framework:** pytest
- **Version Control:** Git and GitHub

Hardware:

Google Colab cloud runtime; computational requirements depend on the selected runtime and corpus size.

